THREE OUTCOMES OF ONE RULE

Dividing two powers of the same base subtracts the exponents. Whether $m - n$ comes out positive, zero or negative decides what the result looks like.

QUOTIENT RULE

$$= a$$

split by m compared with n

requires $a \neq 0$

+

CASE 1

$m > n$

A positive power

a with $m - n > 0$

E.G. $= x$

x^5

1

CASE 2

$m = n$

The result is 1

$a = 1$

−

CASE 3

$m < n$

A negative exponent

a reciprocal; see **negative exponents**

E.G. $= 5 =$

5^{-2} $\frac{1}{25}$

TAKEAWAY

All three cases are natural consequences of subtracting exponents, and all three require $a \neq 0$ and identical bases.